

Complete DevOps & CI/CD Mastery Syllabus

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Program Overview

This syllabus is designed to transform learners into industry-ready:

- DevOps Engineers
- Site Reliability Engineers (SRE)
- Platform Engineers
- Cloud DevOps Engineers
- Infrastructure Engineers
- Kubernetes Administrators
- Automation Engineers
- CI/CD Engineers

The syllabus includes:

- Linux Administration
 - Networking Fundamentals
 - Git & Version Control
 - CI/CD Pipelines
 - Jenkins
 - Docker
 - Kubernetes
 - Terraform
 - Ansible
 - Monitoring & Logging
 - Cloud DevOps
 - Security in DevOps
 - GitOps
 - MLOps Basics
 - Enterprise-Level Projects
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Phase 1 — DevOps Fundamentals

Module 1: Introduction to DevOps

Topics

- What is DevOps?
- Evolution of DevOps
- Traditional IT vs DevOps
- DevOps Lifecycle
- Agile Methodology
- Scrum Basics
- DevOps Culture
- CI/CD Concepts
- Infrastructure as Code
- DevSecOps Overview

Hands-on

- Understanding DevOps workflow
 - DevOps toolchain overview
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Module 2: Linux Fundamentals for DevOps

Topics

- Linux Architecture
- File System Structure
- Linux Commands
- User Management
- File Permissions
- Package Management
- Process Management
- System Monitoring
- Shell Scripting Basics
- Cron Jobs
- SSH & Remote Access

Hands-on

- Linux server administration
 - Shell scripting exercises
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Module 3: Networking Fundamentals

Topics

- OSI Model
- TCP/IP Model
- IP Addressing
- Subnetting
- DNS
- DHCP
- HTTP/HTTPS
- Ports & Protocols
- Firewalls
- NAT
- Load Balancing

Hands-on

- Network troubleshooting
 - DNS configuration
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Phase 2 — Version Control & Collaboration

Module 4: Git & GitHub

Topics

- Version Control Basics
- Git Installation
- Git Commands
- Branching Strategies
- Merge & Rebase
- Pull Requests
- Git Tags
- Git Hooks
- GitHub Workflow
- GitLab Basics

Hands-on

- Git branching workflow
 - Collaborative repositories
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Phase 3 — CI/CD Fundamentals

Module 5: Continuous Integration

Topics

- CI Concepts
- Automated Builds
- Automated Testing
- Build Pipelines
- Artifact Management
- Dependency Management

Hands-on

- CI pipeline creation
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Module 6: Continuous Delivery & Deployment

Topics

- CD Concepts
- Deployment Strategies
- Blue-Green Deployment
- Canary Deployment
- Rolling Updates
- Rollback Strategies
- Release Management

Hands-on

- Automated deployment pipelines
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Phase 4 — Jenkins & Automation

Module 7: Jenkins Fundamentals

Topics

- Jenkins Architecture
- Jenkins Installation
- Jenkins UI
- Jenkins Plugins

- Jenkins Jobs
- Freestyle Projects
- Jenkins Pipelines
- Jenkinsfile
- Pipeline as Code

Hands-on

- Jenkins setup and configuration
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Module 8: Advanced Jenkins

Topics

- Declarative Pipelines
- Scripted Pipelines
- Shared Libraries
- Jenkins Agents
- Distributed Builds
- Credentials Management
- Jenkins Security
- CI/CD Integration

Hands-on

- Enterprise Jenkins pipelines
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Phase 5 — Build & Artifact Management

Module 9: Build Tools

Topics

- Maven
- Gradle
- npm
- pip
- Build Lifecycle
- Dependency Management

Hands-on

- Java and Python project builds
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Module 10: Artifact Repositories

Topics

- Nexus Repository
- JFrog Artifactory
- Docker Registries
- Artifact Versioning
- Package Management

Hands-on

- Artifact storage and management
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Phase 6 — Containerization

Module 11: Docker Fundamentals

Topics

- Containers vs Virtual Machines
- Docker Architecture
- Docker Installation
- Docker Images
- Docker Containers
- Dockerfile
- Docker Volumes
- Docker Networks
- Docker Compose

Hands-on

- Containerizing applications
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Module 12: Advanced Docker

Topics

- Multi-stage Builds
- Docker Swarm Basics
- Docker Security
- Container Optimization
- Image Scanning
- Private Registries

Hands-on

- Production-ready Docker images
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Phase 7 — Kubernetes & Orchestration

Module 13: Kubernetes Fundamentals

Topics

- Kubernetes Architecture
- Pods
- ReplicaSets
- Deployments
- Services
- Namespaces
- ConfigMaps
- Secrets
- Persistent Volumes

Hands-on

- Kubernetes cluster setup
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Module 14: Advanced Kubernetes

Topics

- Ingress Controllers
- Helm Charts
- StatefulSets
- DaemonSets
- Horizontal Pod Autoscaling
- RBAC
- Network Policies
- Kubernetes Security
- Service Mesh Basics

Hands-on

- Production Kubernetes deployment
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Phase 8 — Infrastructure as Code

Module 15: Terraform Fundamentals

Topics

- Terraform Basics
- Providers
- Resources
- Variables
- Outputs
- State Management
- Modules
- Terraform Workflow

Hands-on

- Infrastructure provisioning
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Module 16: Advanced Terraform

Topics

- Remote State
- Workspaces
- Terraform Cloud
- CI/CD Integration
- Multi-cloud Infrastructure

Hands-on

- Enterprise infrastructure automation
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Module 17: Ansible Automation

Topics

- Ansible Architecture
- Inventory Files
- Playbooks
- Roles
- Variables
- Templates
- Ansible Galaxy

Hands-on

- Server configuration automation
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Phase 9 — Cloud DevOps

Module 18: AWS DevOps

Topics

- EC2
- S3
- IAM
- CloudWatch
- CodeBuild
- CodePipeline
- EKS
- ECS
- Lambda

Hands-on

- CI/CD pipeline on AWS
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Module 19: OCI & Azure DevOps Basics

Topics

- OCI DevOps Services
- Azure DevOps Pipelines
- Cloud Deployment Strategies

Hands-on

- Multi-cloud deployments
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Phase 10 — Monitoring & Logging

Module 20: Monitoring Tools

Topics

- Prometheus
- Grafana
- Node Exporter
- Alertmanager
- Monitoring Architecture

Hands-on

- Infrastructure monitoring dashboards
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Module 21: Logging & Observability

Topics

- ELK Stack
- Loki
- Fluentd
- Centralized Logging
- Distributed Tracing
- OpenTelemetry Basics

Hands-on

- Centralized logging system
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Phase 11 — Security & DevSecOps

Module 22: DevSecOps

Topics

- Security in CI/CD
- Secret Management
- Vulnerability Scanning
- SAST
- DAST
- Trivy

- SonarQube
- OWASP Basics

Hands-on

- Secure CI/CD pipelines
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Phase 12 — GitOps & Modern DevOps

Module 23: GitOps

Topics

- GitOps Concepts
- ArgoCD
- FluxCD
- Kubernetes GitOps
- Continuous Reconciliation

Hands-on

- GitOps deployment workflow
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Module 24: Site Reliability Engineering (SRE)

Topics

- SRE Principles
- SLI/SLO/SLA
- Incident Management
- Reliability Engineering
- Chaos Engineering Basics

Hands-on

- Reliability monitoring setup
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Phase 13 — CI/CD for Different Technologies

Module 25: CI/CD for Applications

Topics

- Java CI/CD
- Python CI/CD
- Node.js CI/CD
- React Deployment
- Microservices CI/CD

Hands-on

- Full-stack CI/CD pipelines
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Phase 14 — MLOps Basics

Module 26: MLOps Introduction

Topics

- ML Lifecycle
- ML Deployment
- MLflow
- DVC
- Model Monitoring
- AI Pipeline Automation

Hands-on

- ML deployment pipelines
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Phase 15 — Enterprise Projects

Beginner Projects

- Linux Automation Scripts
- Jenkins CI Pipeline
- Dockerized Web Application
- GitHub Actions Workflow

Intermediate Projects

- Kubernetes Deployment Pipeline
- Terraform Infrastructure Automation
- Monitoring Dashboard
- ELK Logging System

Advanced Projects

- Enterprise CI/CD Platform
 - Multi-cloud Kubernetes Infrastructure
 - GitOps Deployment System
 - AI-powered Monitoring Platform
 - DevSecOps Automation Framework
 - Self-healing Kubernetes Cluster
 - Enterprise Observability Platform
 - Cloud-native DevOps Architecture
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Phase 16 — Interview Preparation

Module 27: DevOps Interview Preparation

Topics

- Linux Interview Questions
- Networking Questions
- Docker Questions
- Kubernetes Questions
- Jenkins Questions
- Terraform Questions
- Scenario-based Troubleshooting
- Architecture Design Questions

Hands-on

- Mock interviews
 - Real-time troubleshooting
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Certifications

Recommended Certifications

- AWS DevOps Engineer
 - Certified Kubernetes Administrator (CKA)
 - Certified Kubernetes Application Developer (CKAD)
 - Docker Certified Associate
 - Terraform Associate
 - Jenkins Certification
 - OCI DevOps Certification
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Recommended Tools

DevOps Tools

- Git
- Jenkins
- Docker
- Kubernetes
- Terraform
- Ansible
- ArgoCD

Monitoring Tools

- Prometheus
- Grafana
- ELK Stack

Cloud Platforms

- AWS
- OCI
- Azure

Programming & Scripting

- Bash
 - Python
 - YAML
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Recommended Learning Timeline

Beginner Level (1–2 Months)

- Linux
- Git
- Networking
- Jenkins Basics

Intermediate Level (3–5 Months)

- Docker
- Kubernetes
- Terraform
- Monitoring

Advanced Level (6–12 Months)

- GitOps
- DevSecOps
- Multi-cloud DevOps
- Enterprise Automation
- SRE Practices

Outcome After Completion

After completing this syllabus, learners can become:

- DevOps Engineer
 - CI/CD Engineer
 - Platform Engineer
 - Cloud DevOps Engineer
 - SRE Engineer
 - Kubernetes Administrator
 - Infrastructure Automation Engineer
 - DevSecOps Engineer
 - Cloud Architect
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Best Learning Strategy

Daily Routine

- 2 Hours Theory
- 3 Hours Hands-on Labs
- 1 Hour Troubleshooting
- 1 Hour Project Building

Weekly Routine

- Deploy one project
 - Automate one workflow
 - Practice Kubernetes
 - Improve CI/CD pipelines
 - Learn troubleshooting scenarios
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Final Capstone Projects

Enterprise-Level Projects

- Enterprise DevOps Platform
 - AI-powered CI/CD Monitoring System
 - Multi-cloud GitOps Platform
 - Kubernetes Auto-healing Infrastructure
 - Intelligent Log Analytics System
 - Secure DevSecOps Pipeline
 - Cloud-native SaaS Deployment Platform
 - Real-time Monitoring & Alerting System
 - Automated Infrastructure Provisioning Platform
 - Enterprise Kubernetes Management System
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Final Advice

To become successful in DevOps & CI/CD:

1. Master Linux and Networking deeply
2. Learn automation consistently
3. Build real-world CI/CD pipelines
4. Practice troubleshooting daily
5. Learn Kubernetes thoroughly

6. Understand cloud infrastructure
7. Focus on Infrastructure as Code
8. Learn monitoring and observability
9. Build enterprise-grade projects
10. Stay updated with modern DevOps practices